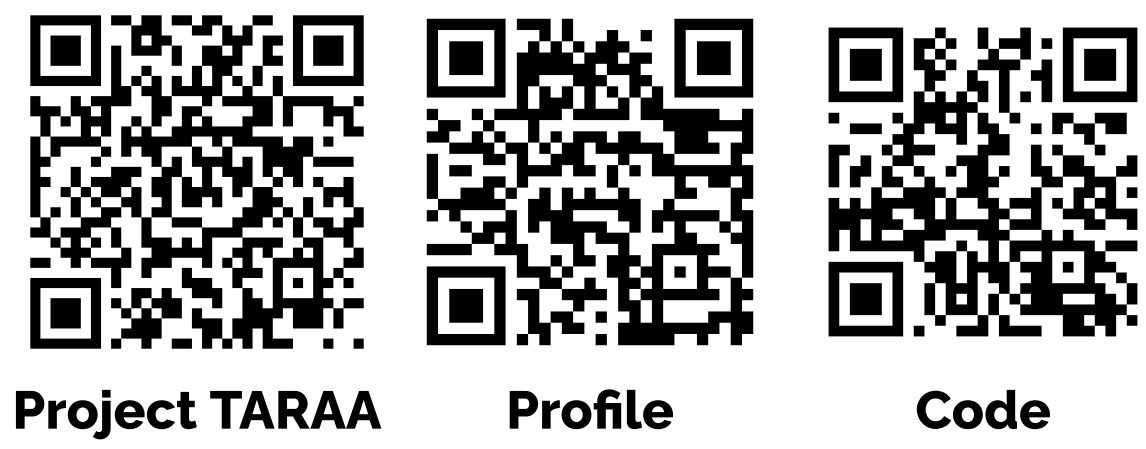


What Do Pathology Foundation Models Learn Beyond Gleason Grade in Prostate Cancer?

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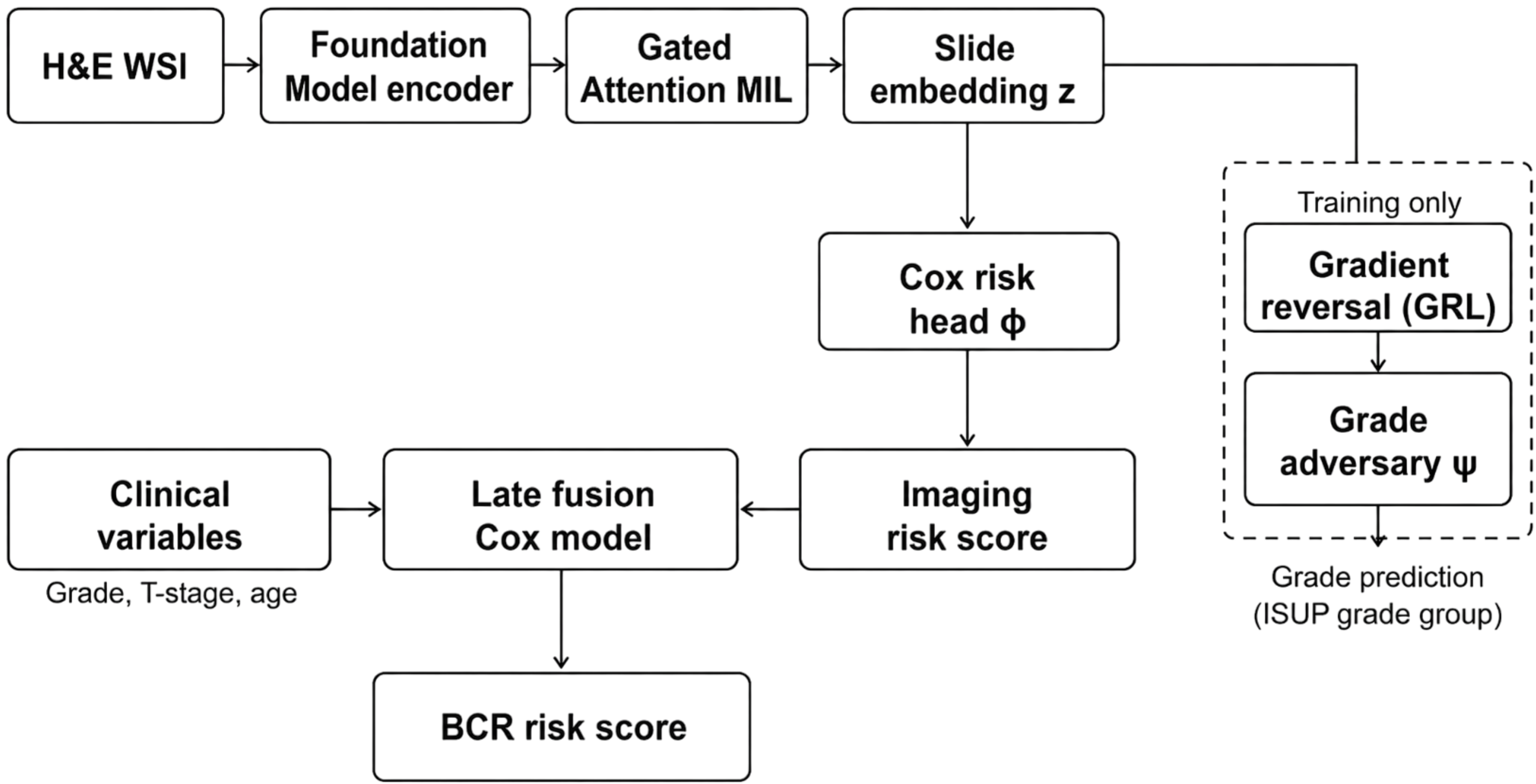
Strasbourg, France

Problem

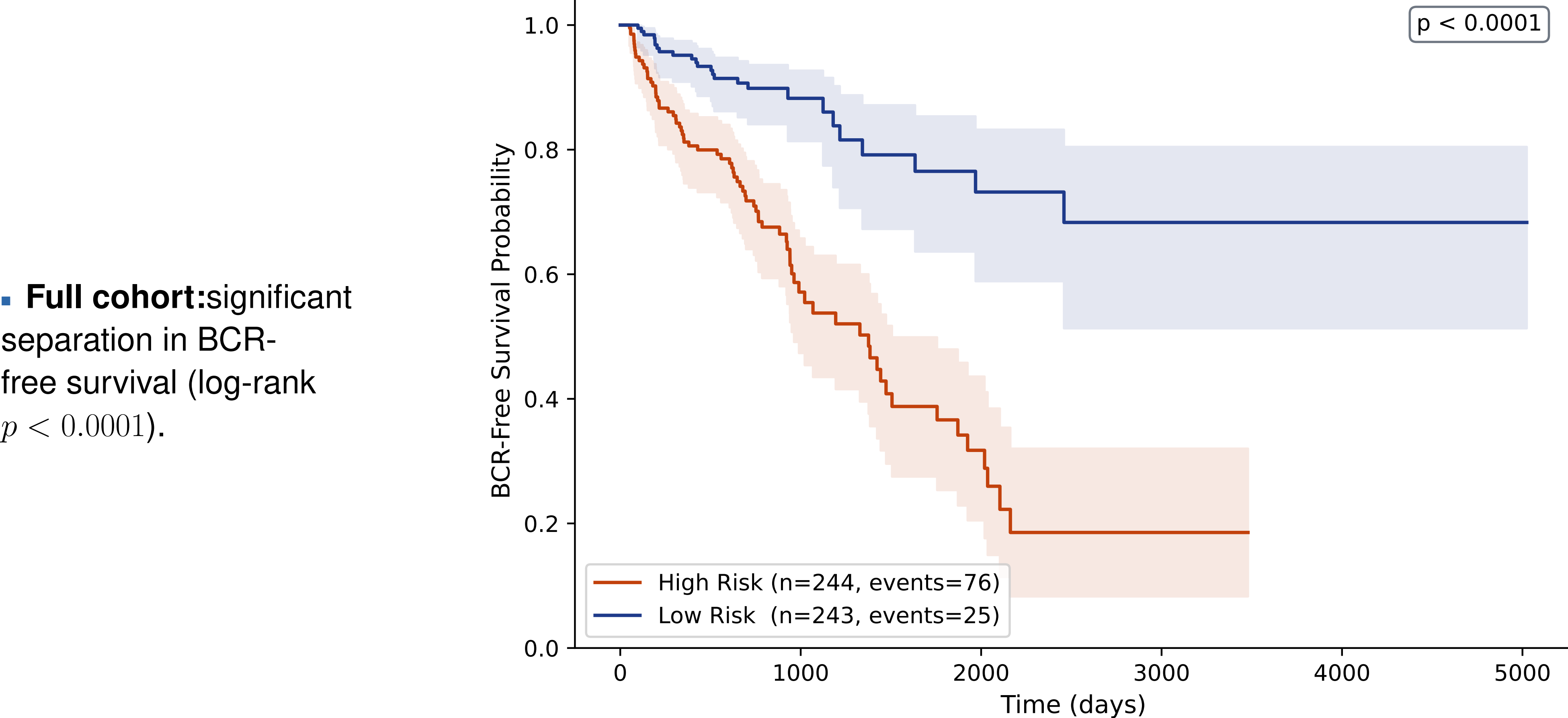
Gleason / ISUP grade is prognostic for biochemical recurrence (BCR) and is reflected in prostate histomorphology, creating a potential source of **grade dependence** in image-based recurrence models.

- Pathology foundation model representations can encode grade-associated morphology from H&E whole-slide images.
- An imaging-derived BCR score may therefore reflect information already captured by Gleason / ISUP grade.
- Predictive discrimination alone does not establish whether the model captures recurrence-relevant morphology **beyond grade**.

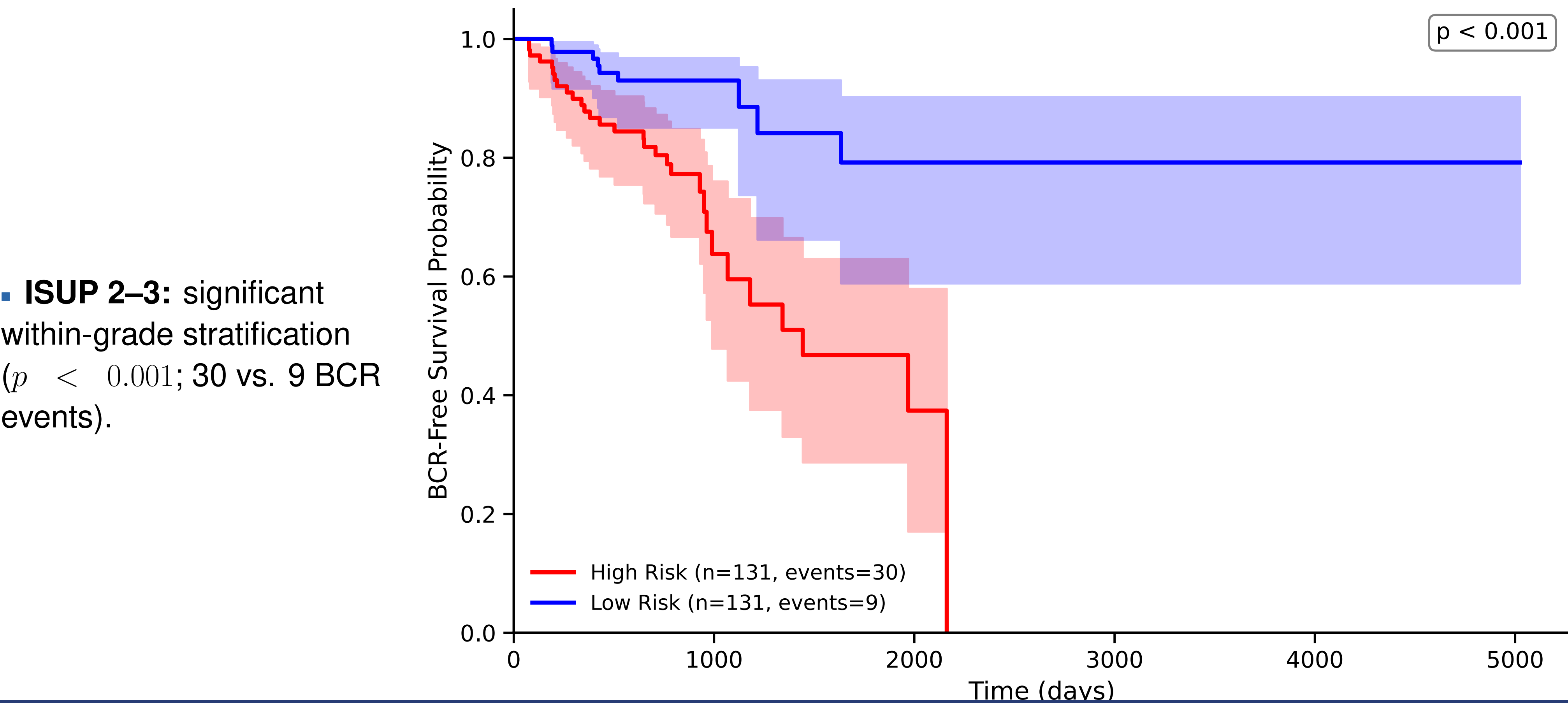
Grade-Disentangled MIL Architecture



BCR-Free Survival: Full TCGA-PRAD Cohort



BCR-Free Survival: Intermediate Grade



- Full cohort:** significant separation in BCR-free survival (log-rank $p < 0.0001$).

- ISUP 2–3:** significant within-grade stratification ($p < 0.001$; 30 vs. 9 BCR events).

Research Objective

Determine whether the pathology foundation-model representations contain prognostic information **for BCR beyond the Gleason / ISUP grade**.

- Quantify the association between imaging-derived recurrence risk and ISUP grade.
- Evaluate whether grade-predictive information can be reduced while retaining recurrence-relevant information.
- Compare grade-disentangled imaging risk with imaging-only, MIL, and clinical baselines.
- Assess risk stratification within grade-defined subgroups, particularly **ISUP 2–3**.

Dataset and Evaluation

TCGA-PRAD (Prostate Adenocarcinoma) with matched H&E-stained whole-slide images and associated clinical variables.

487	101	20.7%	559
patients	BCR events	event rate	median follow-up (days)

Predictive Performance

Method	C-index	95% CI
Clinical Cox (grade, T, age)	0.677	[0.612, 0.726]
ResNet50 + ABMIL	0.566	[0.520, 0.645]
UNI2-h + CLAM	0.627	[0.559, 0.690]
UNI2-h + ABMIL	0.624	[0.549, 0.677]
UNI2-h + TransMIL	0.632	[0.552, 0.682]
UNI2-h + PatchGCN	0.615	[0.556, 0.684]
Virchow2 + ABMIL	0.639	[0.573, 0.704]
UNI2-h + Late Fusion (no GRL)	0.700	[0.633, 0.749]
UNI2-h + GD-MIL	0.704	[0.643, 0.752]
Virchow2 + GD-MIL	0.705	[0.640, 0.756]

- The two GD-MIL encoder variants showed comparable held-out discrimination, indicating limited sensitivity to the choice of foundation-model encoder.

Grade Dependence

Method	C-index	Spearman r
ResNet50 + ABMIL	0.566	0.237
UNI2-h + ABMIL	0.624	0.393
UNI2-h + CLAM	0.627	0.401
UNI2-h + TransMIL	0.632	0.349
UNI2-h + PatchGCN	0.615	0.366
Virchow2 + ABMIL	0.639	0.404
Late Fusion (no GRL)	0.700	0.869
UNI2-h + GD-MIL	0.704	0.340
Virchow2 + GD-MIL	0.705	0.326

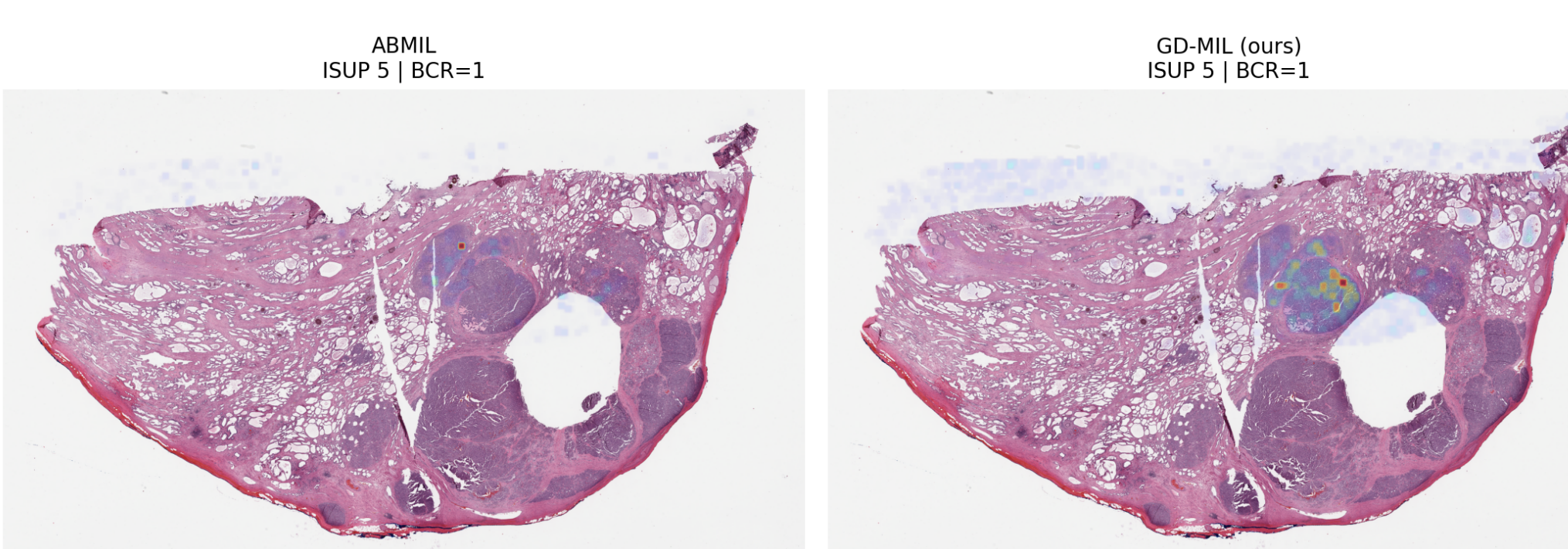
- The no-GRL late-fusion model showed a strong association between predicted recurrence risk and ISUP grade, supporting explicit control of grade information.

Subgroup Analysis

Subgroup	n	Events	C-index	95% CI
Clinically low-risk	244	26	0.650	[0.528, 0.763]
Clinically high-risk	243	75	0.560	[0.487, 0.634]
ISUP 2–3	262	39	0.602	[0.506, 0.693]
ISUP 4–5	183	61	0.574	[0.492, 0.656]

- Discrimination varied across clinical risk groups, and was higher in the clinically low-risk group than in the clinically high-risk group.

Understanding GD-MIL Attention



In a representative **ISUP 5, BCR-positive** case, GD-MIL showed a more focal attention pattern than standard ABMIL after grade disentanglement.

What Did the Foundation Models Learn?

- Their representations encode grade-associated morphology, showing that grade-related structure is embedded in the learned features.
- Recurrence-related variation remains available after reducing grade-predictive information.
- This residual signal is observed within **ISUP 2–3**, indicating prognostic heterogeneity beyond routine histologic grading.